Script-Controlled ACL – Restrict Record Access Based on Field Value
Phase 2.3-Solution Requirements

Date	30 September 2026
Team ID	6ab3ed08a238dc999eb3fda5
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	• Learners register with email or mobile number and password, verified by a one-time password (OTP). • Optional single sign-on with Google or college email. • Password reset via OTP; automatic session timeout after inactivity.	High

2	Authorization levels	• Learner: manage own profile, enroll in courses, take assessments, download and share certificates. • Mentor: view assigned learners, evaluate submissions, answer queries. • Admin: manage users, courses, assessments and reports. • Recruiter: search and view only the profiles and credentials that learners have chosen to share.	High
S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
3	External interfaces	• Payment gateway (e.g., Razorpay) for paid programs. • Email / SMS service for OTPs and notifications. • Google OAuth for sign-in and LinkedIn for one-click credential sharing. • Video / content hosting service for course material.	Medium
4	Transactions processing	• Enrollments, payments, assessment submissions and credential issuance are recorded as time-stamped transactions. • Payments are processed through the gateway with a receipt and automatic retry or rollback on failure. • Each issued credential gets a unique verification ID.	High
5	Reporting	• Learner dashboard showing progress, scores and earned credentials. • Admin dashboard for enrollments, completion rates and assessment results. • Mentor reports on assigned learners; data export in CSV / PDF.	Medium

6	Business rules, etc.	• Certificate is issued only after minimum course completion and a passing score (thresholds configurable by Admin). • Limited number of assessment attempts per course. • Recruiters can view a profile only with the learner's consent.	High
7	Compliance to laws or regulations	• Digital Personal Data Protection Act, 2023: consent, purpose limitation and right to erasure. • IT Act, 2000: reasonable security practices for personal data. • Card data handled only by a PCI DSS-compliant gateway; never stored by the system.	High
S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
8	Other	• Email and in-app notifications for deadlines, results and opportunity alerts. • Learner feedback form and help / support chat.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages, dashboards and course content load quickly; APIs respond promptly; certificates are generated on demand.	• Page load < 2 seconds on a 4G connection • 95% of API calls < 500 ms • Certificate generated < 5 seconds
2	Scalability	The platform handles growth in learners, courses and stored data by scaling out on cloud infrastructure.	• Supports up to 10,000 concurrent users • Handles 5x data growth with no redesign

3	Security & Data Privacy	Personal data and credentials are protected in transit and at rest; access is role-based; vulnerabilities are found and fixed regularly.	• HTTPS (TLS 1.2+) for all traffic • AES-256 encryption at rest; hashed passwords • OWASP Top 10 checks and quarterly vulnerability scans
4	Reliability & Availability	The system stays available and recovers quickly from failures without losing learner data.	• 99.9% uptime per month • Daily backups; recovery within 4 hours
5	Usability & Accessibility	Simple, consistent and mobile-friendly interface that new learners can use without training.	• Complies with WCAG 2.1 AA standards • New learner registers in < 3 minutes • Works on mobile and desktop screens
S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
6	Other	Maintainability and portability: modular, well-tested code that can be deployed on different cloud providers and works across common browsers.	• Automated test coverage of at least 70% • Supports the latest 2 versions of Chrome, Edge, Safari and Firefox